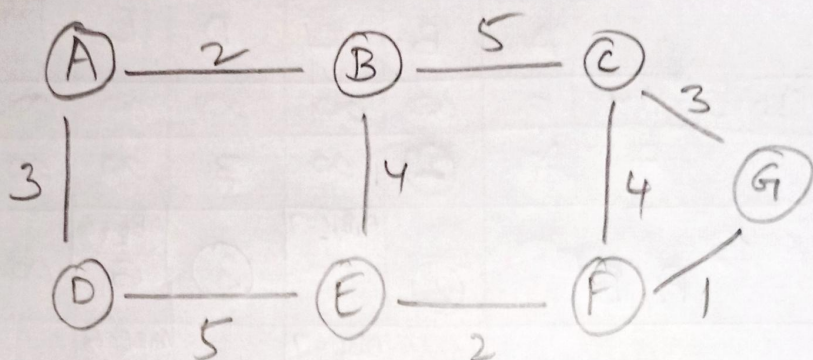


Link State Routing Algorithm



Initial LSDB

	A	B	C	D	E	F	G
A	0	2	∞	3	∞	∞	∞
B	2	0	5	∞	4	∞	∞
C	∞	5	0	∞	∞	4	3
D	3	∞	∞	0	5	∞	∞
E	∞	4	∞	5	0	2	∞
F	∞	∞	4	∞	2	0	1
G	∞	∞	3	∞	∞	1	0

Updated LSDB

Step	A	B	C	D	E	F	G
① Initially		∞	∞	∞	∞	∞	∞
② A		②	∞	3	∞	∞	∞
③ A, B		②	$\overset{A,B,C=7}{7}$	③	$\overset{A,B,E=6}{6}$	∞	∞
④ A, B, D		②	$\overset{A,B,C=7}{7}$	③	$\overset{A,B,E=6, ADE=7}{6}$	∞	∞
⑤ A, B, D, E		②	$\overset{A,B,C=7, ADEBC=17}{7}$	③	⑥	8	∞
⑥ A, B, D, E, C		②	⑦	③	⑥	⑧	9
⑦ A, B, D, E, C, F		②	⑦	③	⑥	⑧	⑨
⑧ A, B, D, E, C, F, G							

* While constructing the LSDB, Parallely the Graph should be constructed.

* If there are any multiple-Path, Consider the Path with Least Cost.

* Here, Circling the Cost means that the link is Confirmed.

* As in this Link State Routing,

Global knowledge of Routing database

is to be Shared with other nodes in

network, the Cumulative-cost of Least-Path is to be Calculated.

Step

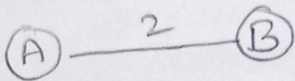
Graph (Least-Cost Path)

After Confirmation

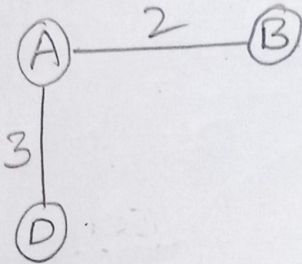
②

①

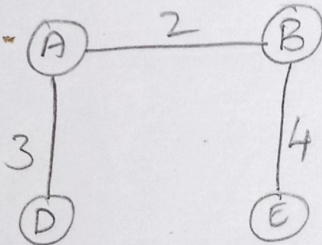
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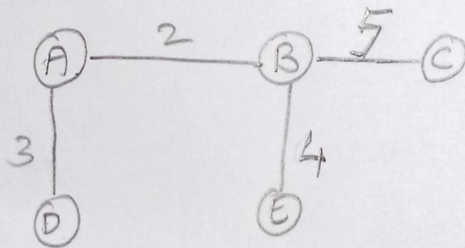
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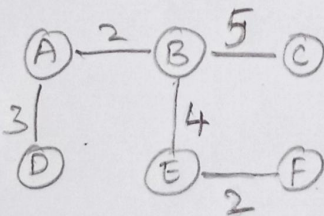
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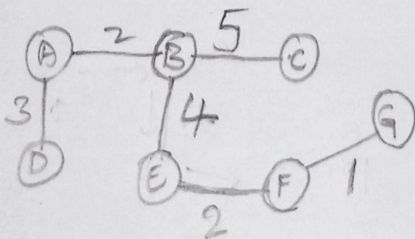
⑥



⑦



⑧



Least cost calculations

From A to A = ①

From A to B = ②
A → B

From A to D = ③
A → D

From A to E

A → B → E = ⑥
A → D → E = 7

From A to C

A → B → C = ⑦
A → D → E → B → C = 17

From A to F

A → B → E → F = ⑧
A → D → E → F = 10
A → B → C → F = 11

From A to G

A → B → C → G = 10
A → B → E → F → G = ⑨